

ZUGSPITZE, Germany

WMO: 109610

Lat: 47.4208N

Lon: 10.9847E

Elev: 2960

StdP: 70.46

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.5	-21.0	-31.4	0.3	-14.5	-28.5	0.4	-12.2	25.2	-10.3	22.6	-9.1	6.0	0	1.095

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	11.0	6.7	9.5	5.6	8.3	4.9	7.8	9.6	6.7	8.4	5.8	7.5	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	9.0	8.2	6.0	8.3	7.3	5.0	7.8	6.3	31.7	9.9	28.9	8.4	26.7	7.6	11.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	18.1	15.0	DB	-25.8	14.3	3.6	1.3	-28.4	15.2	-30.5	16.0	-32.5	16.7	-35.1	17.6	(4)
(5)			WB	-25.9	10.0	3.6	1.0	-28.5	10.7	-30.6	11.3	-32.6	11.9	-35.2	12.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	-10.2	-11.3	-9.9	-7.4	-2.8	0.2	3.6	3.3	1.1	-1.8	-6.6	-8.7
	DBStd	6.76	5.07	5.27	4.61	3.93	3.84	3.84	3.63	3.72	3.86	4.24	4.68	4.77
	HDD10.0	5172	626	597	616	523	396	295	201	209	267	365	497	580
	HDD18.3	8212	884	831	874	773	654	545	458	467	517	623	747	838
	CDD10.0	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	8.5	7.4	7.3	6.7	5.5	5.8	5.5	5.6	6.4	7.0	7.6	8.2
Precipitation	PrecAvg	2008	179	153	190	188	178	183	185	177	132	116	159	177
	PrecMax	2723	440	410	452	554	348	349	381	366	346	302	366	478
	PrecMin	1415	7	30	29	14	20	93	83	61	28	5	0	15
	PrecStd	311	105	85	92	102	81	59	61	60	68	77	83	102
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.8	0.7	3.2	6.8	10.6	13.7	13.2	10.2	8.2	4.2	1.5
		MCWB	-4.3	-4.6	-3.2	-1.5	3.0	6.2	7.0	7.5	4.5	2.1	-2.9	-3.3
	2%	DB	-1.2	-0.7	-1.2	0.9	5.0	8.3	11.3	10.9	8.6	6.2	2.6	0.1
		MCWB	-5.8	-5.1	-4.2	-2.4	2.1	5.1	6.8	6.8	3.7	0.9	-3.4	-4.7
	5%	DB	-2.6	-2.7	-2.4	-0.9	3.3	6.8	9.6	9.5	7.2	4.8	1.2	-1.3
		MCWB	-6.4	-6.0	-4.9	-3.3	1.1	4.3	6.0	5.7	3.1	0.3	-3.8	-4.8
	10%	DB	-4.1	-4.3	-3.8	-2.3	1.8	5.4	8.3	8.1	6.0	3.7	-0.5	-2.7
		MCWB	-6.8	-6.3	-5.6	-3.9	0.2	3.8	5.5	5.0	2.4	-0.3	-4.4	-5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.7	-3.3	-2.3	-0.2	4.7	7.9	9.4	9.4	6.1	3.6	-1.2	-1.2
		MCDB	-1.2	-0.6	-0.7	1.7	6.1	8.9	11.0	10.9	8.3	6.7	0.2	0.0
	2%	WB	-4.2	-4.3	-3.4	-1.7	2.9	6.2	7.9	7.8	5.1	2.2	-2.0	-2.9
		MCDB	-3.1	-2.3	-1.9	0.0	4.3	7.2	9.8	9.8	7.2	4.3	-0.1	-1.7
	5%	WB	-5.5	-5.2	-4.4	-2.6	1.6	5.0	6.9	6.6	4.2	1.4	-2.8	-3.9
		MCDB	-3.9	-3.0	-3.0	-1.4	2.6	6.3	8.6	8.5	6.1	3.5	-0.7	-2.6
	10%	WB	-6.7	-6.2	-5.4	-3.5	0.5	3.8	6.0	5.7	3.4	0.6	-3.5	-5.1
		MCDB	-4.6	-4.4	-4.0	-2.6	1.5	5.1	7.8	7.3	5.0	2.7	-1.1	-3.4
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.1	5.0	5.1	4.9	4.6	4.9	4.8	4.7	4.7	4.7	4.9
		MCDBR	5.7	5.7	5.4	6.1	5.7	5.9	6.4	6.2	5.4	5.1	4.8	5.2
		MCWBR	4.0	3.8	4.5	5.1	4.9	5.7	6.5	6.4	4.7	3.8	3.1	4.2
	5% WB	MCDBR	5.6	5.3	5.4	5.5	5.0	5.4	5.8	5.8	4.9	4.7	4.7	5.3
		MCWBR	4.9	4.5	5.0	4.9	4.8	5.6	6.1	6.1	4.6	4.0	4.1	5.1
Clear-Sky Solar Irradiance	taub	0.193	0.204	0.225	0.247	0.258	0.265	0.265	0.265	0.263	0.247	0.230	0.211	0.192
	taud	2.572	2.555	2.595	2.625	2.643	2.652	2.642	2.667	2.708	2.734	2.690	2.598	
	Ebn at Noon	968	1013	1027	1020	1010	1000	994	990	988	970	938	934	
	Edh at Noon	54	70	80	86	89	89	88	82	71	59	49	47	
All-Sky Solar Radiation	RadAvg	1.34	2.27	3.34	4.27	4.79	5.38	5.34	4.74	3.52	2.43	1.40	1.11	
	RadStd	0.16	0.28	0.37	0.56	0.55	0.55	0.51	0.51	0.43	0.28	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (93 neighbors)	N/A	N/A	N/A	N/A	+0.44	+0.42	N/A	N/A	+56	+13

Nomenclature: See separate page